

Dr. Subrata Karmakar

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Education

- Dec 2025–Present **Research Associate, Centre for Nano Science and Engineering, Indian Institute of Science, Bengaluru, India.**
-Project: "On-Chip Integration of High-Density RRAM Nano-Crossbar Arrays for Ultra-Low-Energy Neuromorphic Accelerators." PI: Dr. Sreetosh Goswami.
- July 2019–Dec 2025 **PhD, Electrical Engineering, Indian Institute of Technology Madras, Chennai, India.**
-Thesis: "Planar Plasmonic Metasurfaces for Phase Manipulation."
Advisor: Prof. Ananth Krishnan, GPA: **8.96/10**.
- Aug 2017–July 2019 **M.Tech. in Optics & Optoelectronics, University of Calcutta, Kolkata, India.**
-Thesis: "Studies on the Optical Characteristics of 1-D Metal-Dielectric Photonic Band Gap Material."
Advisor: Prof. Rajib Chakraborty, % of Marks: **82.62/100**.
- Aug 2009–July 2013 **B.Tech. in Electronics & Instrumentation Engineering, West Bengal University of Technology, Kolkata, India.**
GPA: **8.79/10**

Publications Google Scholar Profile

Journal Articles

- 2026 **Subrata Karmakar**, Anil Ringne, and Ananth Krishnan. Broadband zeroth-order suppression in visible plasmonic metasurfaces via high-period sinusoidal transmissive gratings. *[Manuscript in preparation]*, 2026.
- 2025 Anil Ringne, **Subrata Karmakar**, and Ananth Krishnan. On-axis structured beams generation via moiré and mie resonant metallo-dielectric moiré gratings. *Scientific Reports, Nature*, volume 15, page 16544, May 2025.
- 2024 **Subrata Karmakar**, Anil Ringne, Nirjhar Kumar, and Ananth Krishnan. Uniform dipole resonance and suppressed quadrupole resonance for constant transmittivity full phase control plasmonic metasurfaces. *Scientific Reports, Nature*, volume 14, page 31499, Dec 2024.
- 2024 A. Ringne, N. Kumar, **Subrata Karmakar**, P. Pushkar, and A. Krishnan. Generation of complex beams using flattening of binary gratings. *J. Opt. Soc. Am. B*, volume 41, pages 1364–1372. Optica Publishing Group, May 2024.

Conference Proceedings

- 2018 **Subrata Karmakar**, Rajorshi Bandyopadhyay, and Rajib Chakraborty. Dual tunable multicavity resonator based photonic bandgap structure as hybrid optical filter. In *OSI-International Symposium on Optics(OSI-ISO 2018)*, IIT Kanpur. OSI-International Symposium on Optics(OSI-ISO 2018), 2018.

Industrial Experience

- Aug 2013–June 2017 **Team Leader, Field Engineer**, *Honeywell Automation India Ltd. (Infosoft Digital Design & Services Pvt. Ltd.)*, New Delhi, India.
- 2017 -Project: "Information Technology-Based Building Management System (IT-BMS)," the first of its kind in India, integrating 24 third-party systems through a single interface using different protocols such as BACnet and Modbus.

Skills

Experimental

Nanofabrication: Electron-beam lithography (Raith), electron-beam evaporation (Hind, Techport), sputtering (Techport), photolithography (Heidelberg DWL), dry etching (Oxford), wet-bench operations.

Characterization: Microscopy (Bright-field, Dark-field, Fluorescence, Polarization, Confocal), Profilometer (Bruker), spectroscopic ellipsometer (Woollam), AFM (Park), SEM (Zeiss), FIB (Helios).

Free-space optics: Spatial light modulator (PLUTO-2-VIS-016), spectrometry.

Computational

Electromagnetic simulation: Ansys Lumerical FDTD, Ansys Lumerical mode solver, DDSCAT.

Programming languages: Python, Matlab, Octave.

Fellowships & Awards

- Aug,2019 Awarded the Half-Time Research Assistantship (HTRA) for Ph.D. research at IIT Madras, funded by the Ministry of Education (MoE), Government of India.
- May,2019 University topper in M.Tech. in Optics & Optoelectronics at University of Calcutta, Kolkata, India

Major Teaching Assistantships

National Programme on Technology Enhanced Learning, by the Ministry of Education (MoE), Government of India: Transmission lines and electromagnetic waves, Introduction to Photonics, Optical Fiber Sensors.

IIT Madras course curriculum: Wave Propagation in Communication, Optical Fiber Sensors, Fiber Optic Communication Technology,

Course Work

Introduction to Photonics, VLSI Technology, Wave Propagation in Communication, Advanced Photonics Laboratory, Integrated Optoelectronics Devices and Circuits, Devices and technologies for AI and neuromorphic computing, Optical Engineering, Optical Sensors.

Position Held

2022–2024 I was the official Treasurer of the IIT Madras Student Chapter of OPTICA.

References

Prof. Ananth Krishnan, Professor, Department of Electrical Engineering, IIT Madras, India.
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Prof. Balaji Srinivasan, Professor, Department of Electrical Engineering, IIT Madras, India.
Email: balajis@ee.iitm.ac.in, Contact: +91-44-2257 4426

Declaration

I hereby declare that the information provided in this curriculum vitae is accurate and complete to the best of my knowledge.

Date: August 19, 2026

Place: Bengaluru, India

Subrata Karmakar